

setTimeout aur setInterval

Kisi function ko **abhi nahi, baad me** chalana = "scheduling".

Method	Kaam
<code>setTimeout</code>	function ko ek baar delay ke baad chalata hai
<code>setInterval</code>	function ko baar-baar har interval par chalata hai

Ye JS language ka part nahi, browser aur Node.js dete hain.

setTimeout

javascript

```
let id = setTimeout(func, delay, arg1, arg2);
```

- `delay` milliseconds me (1000 ms = 1 sec), default 0.
- Extra arguments function ko mil jaate hain.

javascript

```
setTimeout(() => console.log("Hi"), 1000);  
setTimeout(sayHi, 1000, "Hello", "John");
```

Galti mat karna: `setTimeout(sayHi(), 1000)` galat hai. Yahan function turant chal jaata hai.
Sahi: `setTimeout(sayHi, 1000)` (bina brackets).

Cancel karna

javascript

```
let id = setTimeout(...);  
clearTimeout(id);
```

setInterval

Syntax same hai. Rokne ke liye `clearInterval(id)`.

javascript

```
let id = setInterval(() => console.log("tick"), 2000);  
setTimeout(() => clearInterval(id), 5000); // 5 sec baad band
```

Nested setTimeout (setInterval ka behtar option)

javascript

```
setTimeout(function tick() {  
  console.log("tick");  
  setTimeout(tick, 2000); // agla call yahin se schedule  
}, 2000);
```

Fayde:

- Agla delay pichhle result ke hisaab se badal sakte ho (jaise server busy ho to delay double kar do).
- Do calls ke beech **minimum delay pakka** rehta hai.
- `setInterval` me function chalne ka time bhi interval me se kat jaata hai, isliye asli gap kam ho sakta hai.

Memory ka dhyan

Scheduled function jab tak chal nahi jaata (ya cancel nahi hota), memory me rehta hai, aur uske outer variables bhi. Zarurat na ho to cancel kar do.

Zero delay: `setTimeout(func, 0)`

Function **current script khatam hone ke baad** jaldi se chalta hai.

javascript

```
setTimeout(() => console.log("World"));  
console.log("Hello");  
// Hello, phir World
```

Browser me nested timers 5 baar ke baad kam se kam 4ms delay ke saath chalte hain (purani wajah se).

Yaad rakho

- Timers **exact time guarantee nahi** karte. CPU busy ho, tab background me ho, ya battery saver ho to der ho sakti hai.
- `setTimeout` ka function tabhi chalta hai jab baaki code ka kaam ho jaaye. Heavy loop chal raha ho to timer ko wait karna padta hai.